

Ian Zhang

Curriculum Vitae

Contact Information

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Education

- MSc in Statistics**, University of Toronto *2025-2026*
- Advisor: Nancy Reid
 - Relevant coursework: Applied statistics, mathematical statistics, graduate probability, survival analysis, asymptotic statistics, statistical inference
- Honours BSc in Statistics/Mathematics**, University of Toronto *2021-2025*
- Advisor: Jun Young Park
 - Graduated with high distinction
 - Relevant coursework: Machine learning, stochastic processes, statistical inference, methods of multivariate data, real analysis, linear algebra, ordinary and partial differential equations, introduction to computer science and programming

Publications and Preprints

†: Corresponding Author; *: Equal contribution

Under Review

1. Michael A. Kouritzin, **Ian Zhang**[†], Jyoti Bhadana, Seoyeon Park *2026*
Markov Processes for Enhanced Deepfake Generation and Detection
<https://arxiv.org/abs/2411.07993>

In Progress

1. Aditya Khan, **Ian Zhang** *2025*
Parametric Hotspot Analysis with the Getis-Ord Statistic
2. Hanlong Chen*, **Ian Zhang***, Bruna Seixas-Lima, Howard Chertkow, ..., Malcolm Binns *2026*
Reproducible Dementia Subtyping through Cross-Validated Joint Hyperparameter Optimization of PCA dimensionality and k in Unsupervised Clustering
3. **Ian Zhang**, Thibault Randrianarisoa *2026*
Likelihood tempering to mitigate prior dominance in variational posteriors for Bayesian neural networks

Research Experience

- Graduate Research Assistant**, Baycrest Institute/University of Toronto *2025-Present*
Supervisor: Malcolm Binns
- Summer Research Trainee**, McGill University *2025*
Supervisor: Jianguo (Jeff) Xia
- Research Assistant**, University of Alberta *2023-2024*
Supervisor: Mike Kouritzin

Teaching

Teaching Assistant

MAT137: Calculus with Proofs

Fall 2025 & Winter 2026

STA302: Methods of Data Analysis I

Fall 2025

STA437: Methods of Multivariate Data

Winter 2026

Presentations

Talks

2025 SSC Annual Meeting in Saskatoon

2025

Presented *Markov Processes for Enhanced Deepfake Generation and Detection* in the Probability session.

Third Joint SIAM/CAIMS Annual Meetings

2025

Presented *Markov Processes for Enhanced Deepfake Generation and Detection* in the Machine Learning and Stochastic Processes session.

Posters

The Thirteenth Annual Canadian Statistics Student Conference

2025

Presented *Comparing Exact and Approximate Inference for Spatial Autocorrelation Tests: A Power Analysis*.

Honours and Awards

Anna & Alex Beverly Memorial Fellowship

2025

Awarded by University College (University of Toronto) for general proficiency during my undergraduate degree.

NSERC Undergraduate Student Research Award (x2)

2023, 2024

Awarded by the University of Alberta to research Markov process methods for deepfake generation and detection.

Jack Zwaigenbaum Scholarship

2022

Awarded by University College (University of Toronto) for academic excellence in my first year of undergraduate studies.

Dean's List Scholar (x4)

2022, 2023, 2024, 2025

Awarded by the Faculty of Arts and Science (University of Toronto) for maintaining academic excellence every year of my undergraduate degree.

Grants

CSSC Travel Grant

2025

Awarded to full-time Canadian students presenting a poster or giving a talk at the Canadian Statistics Student Conference.

Services

Conferences

Session chair, 2025 Third Joint SIAM/CAIMS Annual Meetings

Chaired the “Machine Learning and Stochastic Processes” session.

Other

Citizenship: Canadian

Languages Spoken: English (native), Mandarin (native), Japanese (intermediate)

Memberships: Statistical Society of Canada